

Cadena de Markov

Definición 1 (Cadena de Markov). Sean $\forall n \in \mathbb{N} : X_n : \Omega \rightarrow \mathbb{N}$ variables aleatorias discretas, $(X_n)_{n=0}^{\infty}$ es una cadena de Markov $\iff \forall n \in \mathbb{N} \cup \{0\} : \forall i, j \in \mathbb{N} :$

$$\mathbb{P}(X_{n+1} = j \mid X_n = i \wedge \dots \wedge X_0 = i_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

Es decir, la probabilidad de que la variable aleatoria X_{n+1} tome el valor j condicionada a todos los valores anteriores de la cadena, solo depende del valor de la variable aleatoria X_n .

Referenciado en

- [cadena-markov-recurrencia](#)
- [cadena-markov-transitoriedad](#)
- [cadena-markov-comunicacion](#)
- [cadena-markov-irreducible](#)
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